

4) Suppose that the data set analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 45, 46, 52, 70.

(a) What is the mean of the data? What is the median.

$$A) \text{ mean } (\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 45 + 46 + 52 + 70}{27}$$

$$= \frac{809}{27} = 29.96 \approx 30$$

$$\text{median} = \frac{n+1}{2} = 14^{\text{th}} \text{ value}$$

$$= \frac{27+1}{2} = \frac{28}{2} = 14$$

$$14^{\text{th}} \text{ value} = 25$$

$$\text{median} = 25$$

(b) What is the mode of the data? Comment on the data's modality.

A) 25, 35 are Bimodal

3) (c) midrange of data?

$$A) \text{ midrange} = \frac{13 + 70}{2} = \frac{83}{2} = 41.5$$

$$(\text{midrange} = 41.5)$$

(d) First quartile (Q_1) & Third quartile (Q_3)

A) $Q_1 = 13, 19, 16, 16, 19, 20, 20, 21, 22, 22, 25, 29, 25$

middle value = 20

$$Q_1 = 20$$

$Q_3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70$

middle value = 35

$$Q_3 = 35$$

$$IQR = 35 - 20 = 15$$

(e) Five-Number Summary of the data.

A) minimum = 13

$$Q_1 = 20$$

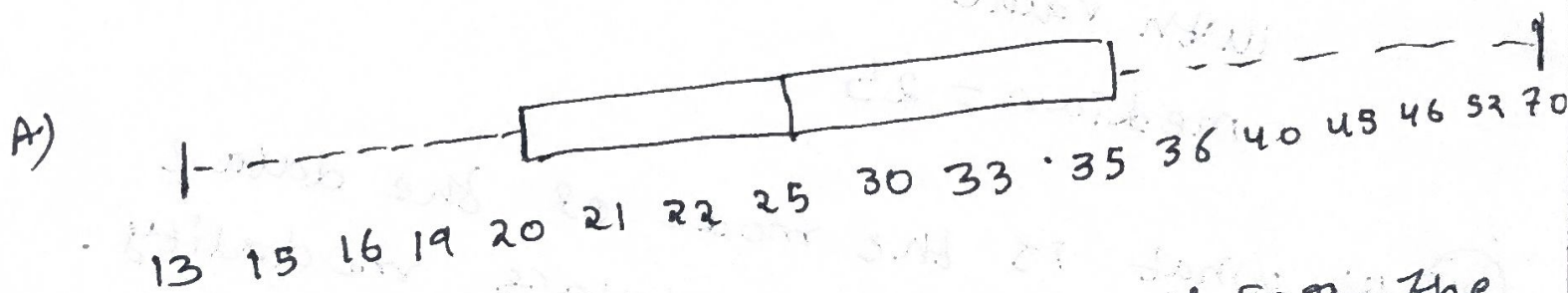
median = 25

$$Q_3 = 35$$

maximum = 70

(13, 20, 25, 35, 70)

(f) ~~Boxplot~~ Show a boxplot of the data.



(2) Given the following measurements for the variable age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28, standardize the variable by the following: compute the mean absolute deviation of age.

A)

$$\text{mean} (\bar{x}) = \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28}{10} = \frac{330}{10}$$

$$\text{mean} = 33$$

Age(x)	18	22	25	42	28	43	33	35	28
$ x - \bar{x} $	15	11	8	9	5	10	0	2	5

$$\text{mean Absolute deviation (MAD)} = \frac{\sum |x - \bar{x}|}{n} = \frac{88}{10} = 8.8$$

$$\therefore \text{mean} = 33$$

$$\text{MAD} = 8.8$$

③ Suppose the data for analysis includes the attribute age. The age values for the data samples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means smooth the above data, using a bin depth of 3. ~~I~~ Illustrate your steps

A) Step 1:-

Total values = 27

Bin depth = 3

$$\text{Total Bins} = \frac{\text{Total values}}{\text{Bin depth}} = \frac{27}{3} = 9$$

Bin	Original values	Bin means	Smoothed values
B ₁	13, 15, 16	14.67	14.67, 14.67, 14.67
B ₂	16, 19, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21	21, 21, 21
B ₄	22, 25, 25	24	24, 24, 24
B ₅	25, 25, 30	26.67	26.67, 26.67, 26.67
B ₆	33, 33, 35	33.67	33.67, 33.67, 33.67
B ₇	36, 40, 45	35	35, 35, 35
B ₈	36, 40, 45	40.33	40.33, 40.33, 40.33
B ₉	46, 52, 70	56	56, 56, 56

④ Use the two methods below to normalize the following group of data: - 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 & max = 1 (b) z-score normalization.

A) a) min-max normalization (min = 0, max = 1)

$$\text{Normalized value} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

minimum = 200, maximum = 1000

Original value	Normalized value
200	$(200 - 200) / 800 = 0$
300	$(300 - 200) / 800 = 0.125$
400	$(400 - 200) / 800 = 0.25$
600	$(600 - 200) / 800 = 0.5$
1000	$(1000 - 200) / 800 = 1$

(0.00, 0.125, 0.25, 0.50, 1)

b) z-score normalization

$$z = \frac{x - \mu}{\sigma}$$

$$\text{mean } (\mu) = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{(-300)^2 + (-200)^2 + (-100)^2 + 100^2 + 500^2}{5}} = \sqrt{80000} \approx 282.84$$

Original value

200

300

400

600

1000

-1.06

00,
n=20

Original Value	Z-Score
200	$(200-500)/282.84 = -1.06$
300	$(300-500)/282.84 = -0.71$
400	$(400-500)/282.84 = -0.35$
600	$(600-500)/282.84 = 0.35$
1000	$(1000-500)/282.84 = 1.77$

-1.06, -0.71, -0.35, 0.35, 1.77